

HAMBURG FUHLSBUTTEL, Germany

WMO: 101470

Lat: **53.6331N**Lon: **9.9881E**Elev: **49**StdP: **14.67**Time Zone: **1.00 (EUC)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	21.0	10.3	9.3	19.7	15.4	12.0	23.4	27.3	45.0	24.9	44.9	7.9	100	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	84.5	66.6	80.7	66.1	77.2	64.2	69.2	79.9	67.5	77.4	65.8	74.3	9.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	96.0	72.7	64.3	90.4	71.1	62.6	85.1	69.5	33.2	80.3	31.9	77.3	30.7	74.5	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.9	17.9	DB	11.9	91.7	5.9	4.3	7.7	94.8	4.2	97.3	0.9	99.8	-3.5	102.9	(4)
(5)			WB	11.3	73.0	6.0	1.3	6.9	74.0	3.4	74.8	0.0	75.5	-4.4	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	34.9	35.6	40.8	49.3	55.3	61.2	65.6	64.0	58.9	51.3	43.2	38.6	(6)	
	DBStd	12.53	7.90	6.77	7.60	6.74	6.90	5.79	5.96	5.02	5.10	5.95	6.67	7.50	(7)	
	HDD50	1911	467	404	292	94	24	0	0	0	1	56	217	356	(8)	
	HDD65	5668	932	824	750	473	310	144	65	79	193	426	653	819	(9)	
	CDD50	1900	0	0	7	71	189	335	484	433	270	95	14	2	(10)	
	CDD65	182	0	0	0	0	10	29	85	47	11	0	0	0	(11)	
	CDH74	1573	0	0	0	10	102	277	757	351	72	4	0	0	(12)	
	CDH80	422	0	0	0	0	18	60	243	93	8	0	0	0	(13)	
(14) Wind	WSAvg	9.0	10.4	9.8	9.8	9.1	8.9	8.4	8.1	7.7	8.1	8.6	9.2	10.3	(14)	
(15) Precipitation	PrecAvg	30.5	2.5	1.8	2.2	1.8	2.3	3.0	3.2	3.0	2.6	2.4	2.7	2.9	(15)	
	PrecMax	42.2	5.3	5.2	5.9	4.5	6.3	8.3	6.9	9.3	7.1	7.2	5.7	6.9	(16)	
	PrecMin	19.3	0.0	0.2	0.2	0.0	0.2	0.4	0.2	0.3	0.4	0.6	0.3	0.5	(17)	
	PrecStd	5.0	1.3	1.1	1.3	1.0	1.2	1.6	1.7	1.5	1.4	1.3	1.3	1.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	55.2	64.5	75.3	82.3	85.7	91.6	88.0	80.9	71.6	60.8	55.5	(19)	
		MCWB	50.4	45.3	51.8	57.8	65.2	67.1	69.7	68.5	66.7	61.6	56.4	51.8	(20)	
	2%	DB	50.3	50.0	59.0	70.0	76.7	81.0	86.1	82.1	75.5	66.1	57.3	53.2	(21)	
		MCWB	48.2	46.5	50.2	55.6	62.5	66.4	67.4	66.7	64.0	59.9	54.4	50.4	(22)	
	5%	DB	48.1	46.7	54.0	66.2	72.0	76.9	82.2	77.2	71.7	62.5	55.0	50.3	(23)	
		MCWB	46.6	44.1	48.0	54.4	59.0	63.7	66.3	64.5	62.2	57.8	52.1	48.3	(24)	
	10%	DB	46.1	44.7	51.5	62.2	68.2	73.0	78.4	73.7	67.9	59.3	52.0	48.1	(25)	
		MCWB	44.6	42.4	46.5	52.7	58.1	61.8	65.3	63.1	60.5	55.7	49.3	46.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	48.9	54.6	59.6	67.5	70.3	71.9	70.7	67.5	62.8	57.9	53.0	(27)	
		MCDB	53.1	50.1	60.9	71.1	79.6	80.3	83.9	81.1	76.5	69.3	60.2	55.2	(28)	
	2%	WB	48.7	46.4	51.4	57.0	64.0	67.6	69.5	68.3	65.3	60.1	54.6	50.9	(29)	
		MCDB	50.1	49.0	56.0	67.5	72.8	78.3	81.1	78.4	73.5	64.8	57.0	52.6	(30)	
	5%	WB	46.3	44.3	49.3	55.3	61.3	65.3	67.9	66.6	63.2	58.2	52.4	48.7	(31)	
		MCDB	47.7	46.6	53.8	63.6	69.9	73.6	78.2	74.9	69.8	61.9	54.8	50.3	(32)	
	10%	WB	44.0	42.2	46.8	53.1	58.8	63.4	66.3	64.8	61.6	56.3	50.1	46.0	(33)	
		MCDB	45.2	44.2	50.6	61.2	65.9	71.0	75.3	71.8	67.0	59.5	51.8	47.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.9	12.8	16.8	17.0	17.1	16.7	15.9	14.4	11.8	8.5	7.4	(35)	
		MCDBR	8.4	11.7	18.8	24.2	23.6	24.2	24.3	21.8	20.5	15.4	9.7	8.8	(36)	
	5% WB	MCWBR	8.1	9.0	11.8	12.6	12.0	12.1	9.6	9.9	10.9	9.9	7.8	7.6	(37)	
		MCWBR	7.9	11.1	16.6	21.4	21.0	20.8	20.6	18.6	17.2	13.8	9.5	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.318	0.337	0.367	0.419	0.412	0.409	0.428	0.421	0.392	0.356	0.334	0.310	(40)	
		taud	2.344	2.348	2.290	2.180	2.243	2.277	2.240	2.272	2.336	2.427	2.390	2.339	(41)	
	Ebn at Noon		204	240	256	256	265	267	259	253	245	231	197	183	(42)	
		Edh at Noon	18	25	32	41	41	40	41	37	31	23	18	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		178	392	771	1276	1603	1681	1591	1317	958	527	223	134	(44)	
	RadStd		21	54	77	162	186	131	216	99	109	61	23	9	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page